

TASK 1: Java Environment Setup & First Executable Program

Tools:

- Primary: JDK 8+ (OpenJDK), Command Prompt / Terminal
- Alternatives: IntelliJ IDEA Community, Eclipse IDE, VS Code + Java Extension Pack

Hints / Mini Guide:

1. Download and install OpenJDK from the official OpenJDK source and properly configure the JAVA_HOME environment variable to ensure Java commands work globally across the system.
2. Verify Java installation by running `java -version` and `javac -version` commands to confirm both runtime and compiler are functioning correctly.
3. Create a new directory structure manually to understand how Java projects are organized at a basic filesystem level.
4. Write a simple Java class using a text editor and define a `main()` method following correct Java syntax rules.
5. Compile the Java file using the `javac` compiler and understand how `.class` bytecode is generated.
6. Execute the compiled program using the `java` command and observe output in the terminal.
7. Modify the program to accept command-line arguments and print them, reinforcing how Java applications receive runtime input.
8. Introduce basic comments and formatting standards to align with professional coding practices.

Deliverables:

- Compiled and executed Java program
- Screenshot or log of terminal execution
- Source code file with comments

Interview Questions Related To Above Task:

- What is the role of JVM, JRE, and JDK?
- Why is `main()` static in Java?
- Difference between compilation and execution?
- What is bytecode?
- How does Java achieve platform independence?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

